

Transform Array In-Place

Difficulty: Medium | Accuracy: 56.34% | Submissions: 138K+ | Points: 4

Given an integer array `arr[]` of size `n`, where each element lies in the range `[0, n-1]`, transform the array such that every element at index `i` becomes:

$$\text{arr}[i] = \text{arr}[\text{arr}[i]]$$

The transformation must be performed in-place, meaning the same array should store both the original and updated values without using another array.

Examples

Input: `arr[] = [1, 0]`

Output: `[0, 1]`

Explanation: The original array is `[1, 0]`. At index 0, `arr[0] = 1`, so the new value becomes `arr[1] = 0`. At index 1, `arr[1] = 0`, so the new value becomes `arr[0] = 1`. Thus, the transformed array becomes `[0, 1]`.

Input: `arr[] = [4, 0, 2, 1, 3]`

Output: `[3, 4, 2, 0, 1]`

Explanation: The original array is `[4, 0, 2, 1, 3]`. Each element is replaced by the value at index `arr[i]` in the original array. So the new values become 3, 4, 2, 0, 1 respectively, and the transformed array becomes `[3, 4, 2, 0, 1]`.

Constraints

$$1 \leq n \leq 4 \cdot 10^4$$
$$0 \leq \text{arr}[i] < n$$

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